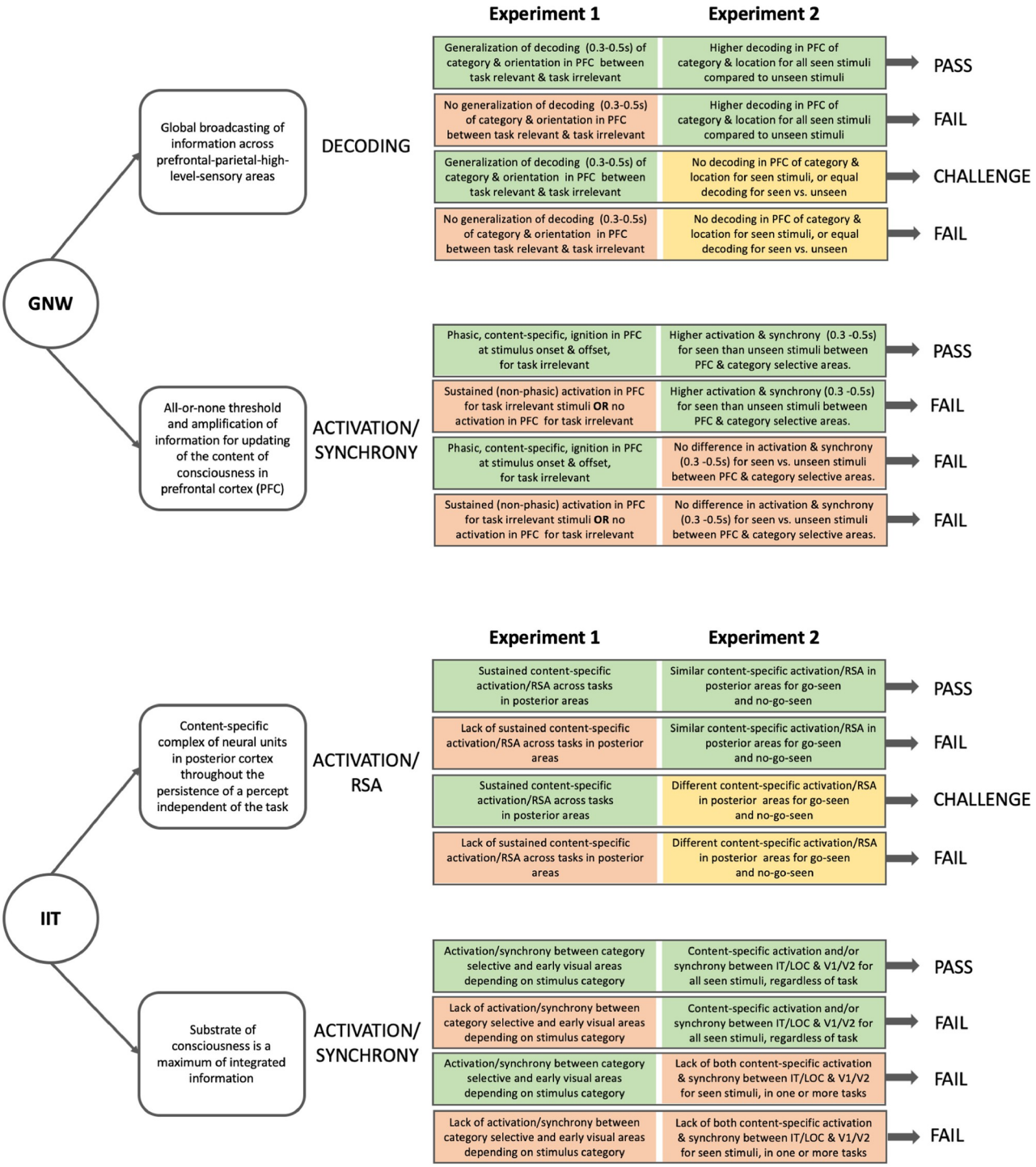


**Fig 1. Core predictions, analyses, and potential results for the two experiments.** For each theory (white circles), the core predictions (white rectangles) and possible outcomes (colored rectangles) are summarized. Green squares denote positive outcomes, red ones denote negative outcomes, and yellow squares denote negative outcomes that are less conclusive, as they might stem from limitations of the methods or analyses. On the right, we conclude how combinations of results would reflect on the theories. Note that some predictions require time-resolved measures (M-EEG, iEEG), while others can be tested with all three techniques. PFC refers to Prefrontal cortex; RSA refers to representational similarity analysis [RSA; 48]; IT refers to inferotemporal cortex; LOC refers to lateral occipital cortex; V1 refers to primary visual cortex.

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methods. This will allow for the development of analyses while still committing to a preregistered protocol. The data will be split into two halves, so the first-half is used to optimize the analysis methods (phase 2, analysis optimization), with the aid of expert advisors. Then, analysis routines and all specific details will be locked, the preregistration will be amended, and the second half of the data will be used for replication (phase 3, replication). Thus, in accordance with our plan, the current protocol includes a higher-level description of the analyses and explicit, fleshed-out predictions, but does not include all of the analysis details.

**Statistical approach.** We plan to use a combination of classical and Bayesian statistics. The advantages of using Bayesian inference are threefold. First, it will enable us to assess the weight of evidence for each theories' prediction for the different analyses. Second, it will allow us to conclude that there is no evidence supporting a prediction (or for one theory over the other) when the data fails to show a predicted effect. This is a key point that contrasts with classical inference, where the null hypothesis of no difference can never be accepted. Third, it will allow us to assess the likelihood of the theories across the different analyses and different brain recording methods.

## Materials and methods

### Status and timeline

In early 2018, a group convened at the Allen Institute for Brain Sciences to map out contrasting predictions and ideas for experiments and methods. Over the course of the following months, the three first authors worked in conjunction with the leading theorists, and with input of the group, to specify experiments, instrumentation, recording methods, sample sizes, exclusion criteria, and analysis methods, which were all preregistered (https://osf.io/mbcfy/). During 2019, the three first authors carried out pilot behavioral and eye-tracking studies to refine and finalize the experimental designs. The initial preregistration was accordingly amended. While deploying and pilot testing the experiments in the six data acquisition labs, additional changes were introduced, and were again preregistered. The original and the amended preregistration files were frozen prior to the commencement of data acquisition.

The project is tripartite. At the time of submission of this study protocol, phase 1 is approximately halfway complete. In **phase 1**, all data will be acquired by theory neutral teams. To ensure replicability of the results, the entire dataset will be split into two halves, each with an equal mixture of data from each of the two labs for each recording technique (half of the participants tested in both labs). In **phase 2**, after evaluating data quality (see OSF preregistration), the first half of the data will be used for developing analysis tools (optimization of methods). The purpose of **phase 2** is to define the best analysis practices and to agree upon, in consultation with expert advisors, the detailed analysis protocols that will be preregistered in the final, unchangeable version of the preregistration. In our original protocol, phase 1 and phase 2 were planned to be conducted successively, yet due to delays in data acquisition caused by the COVID-19 pandemic of 2020–2021, now phase 1 and phase 2 will overlap. In **phase 3**, the replication phase, the second half of the data will be analyzed using these preregistered, agreed upon protocols, thereby allowing an in-house replication of the results obtained in phase 2. Data will be analyzed by the acquiring laboratories, a data analysis team composed of the two theory leaders, the center PIs (the three co-first authors leading this project), and a method-experts advisory panel. All data, meta-data, stimulation protocols, and analysis pipelines will be publicly released upon the publication of the final results via a combination of documents in OSF, raw data, processed data and analysis code, and protocol instructions and specifications in SLAB (data management and sharing plan are detailed below).

## Experiment 1

**Aim.** Experiment 1 will test several predictions of GNW and IIT (detailed below): the most critical for GNW relates to its predictions concerning information about a clearly visible stimulus, the decodability of which should not depend on whether the percept is task-relevant or not; and IIT's critical prediction that the physical substrate of consciousness should remain active throughout the presentation of a clearly visible stimulus, whether task-relevant or not.

**Design, stimuli, and procedure.** We will measure brain activity elicited by visual stimuli that are clearly consciously perceived while manipulating (1) the task, such that some stimuli are task-relevant while others are task-irrelevant [following 49], (2), stimulus duration, such that stimuli are perceived for different durations [50], (3) stimulus category, such that the content of perception varies, and (4) stimulus orientation, such that the specific features of the perceived content within a given category can be tested under fully task-irrelevant conditions.

We will use four categories of stimuli that naturally fall into two distinct classes—pictures (20 faces, half male, half female, and 20 objects) and symbols (20 letters and 20 false-fonts). Stimulus orientation will be manipulated, *i.e.*, half will have a side view (rotated +/- 30˚) and half a front view. All stimuli will be supra-threshold, presented at fixation and subtend the same size in visual angle (6˚x6˚).

In each 2s trial, one stimulus from one of the four categories will be presented, in greyscale, for a duration of 500, 1000 or 1500ms, followed by a blank. The experiment will be divided into mini-blocks, so that for each mini-block, half the trials will contain task-relevant stimuli and the other half task-irrelevant stimuli. To define task-relevance, subjects will be instructed to detect the occurrences of two targets regardless of their orientations (targets in both orientations will be presented at the beginning of the trial) belonging to two different categories (counterbalanced between mini-blocks) (Fig 2). Accordingly, each mini-block will contain three different trial types: i) *Task-Relevant Targets*: the two stimuli being detected in that block (e.g., a specific face and object; upper row in Fig 2; or a specific letter and a false-font; lower row in Fig 2); ii) *Task-Relevant Non-Targets*: stimuli from the task-relevant categories but not the specific targets (e.g., other faces and objects; highlighted in blue in upper row in Fig 2); and iii) *Task-Irrelevant Stimuli*: stimuli from the other categories (e.g., letters and false-fonts; highlighted in green in upper row in Fig 2). Thus, the very same stimuli will be task-relevant non-targets in some blocks and task-irrelevant in other blocks. Subjects will be asked to maintain central fixation throughout each trial. Gaze will be monitored online through an eye tracker.

Thus, in formal terms, Experiment 1 is a nested factorial design with four factors: stimulus category (face, object, letter, false-font), task relevance (task-relevant targets, task-relevant non-targets, task-irrelevant stimuli), stimulus duration (500, 1000, 1500 ms), and stimulus orientation (front view, side view). We are particularly interested in the interactions between stimulus category and task-relevance, as well as between stimulus duration and task-relevance. Trial numbers, stimulus randomization and timing parameters are optimized for the different methodologies i.e., fMRI, M-EEG and iEEG. Technique specific details are described in the preregistration in OSF.

**Participants, sample size, exclusion criteria and stopping rule.** For the fMRI and M-EEG studies, all participants will be older than 18 years old, have reportedly normal or corrected-to-normal visual acuity and no known history of psychiatric or neurological disorders. For the iEEG studies, subjects will be 10–65 years old, be able to provide informed consent, have IQ $> 70$, be fluent in English or Spanish, self-reported normal hearing, normal or corrected-to-normal vision, and show cognitive and language abilities within or above the normal range in formal neuropsychological testing performed before surgery and must not have had